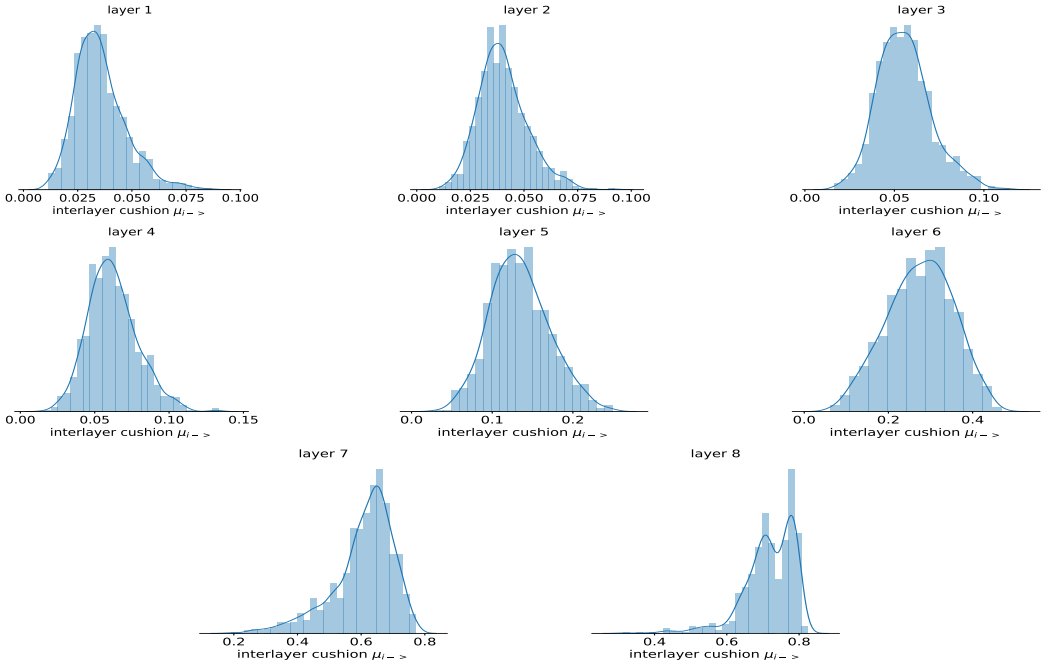
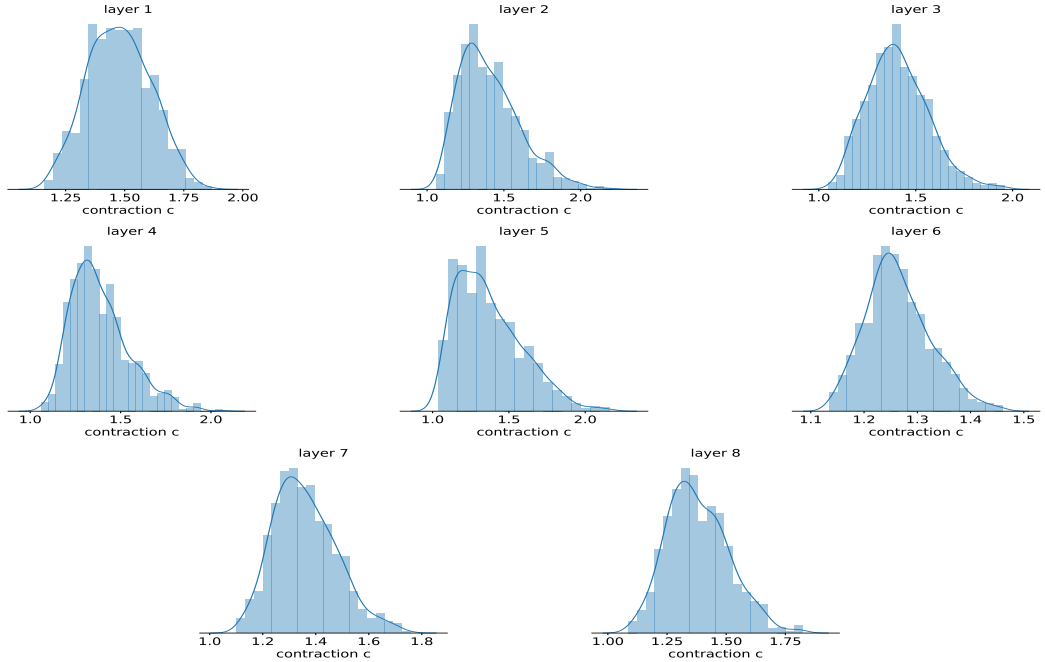
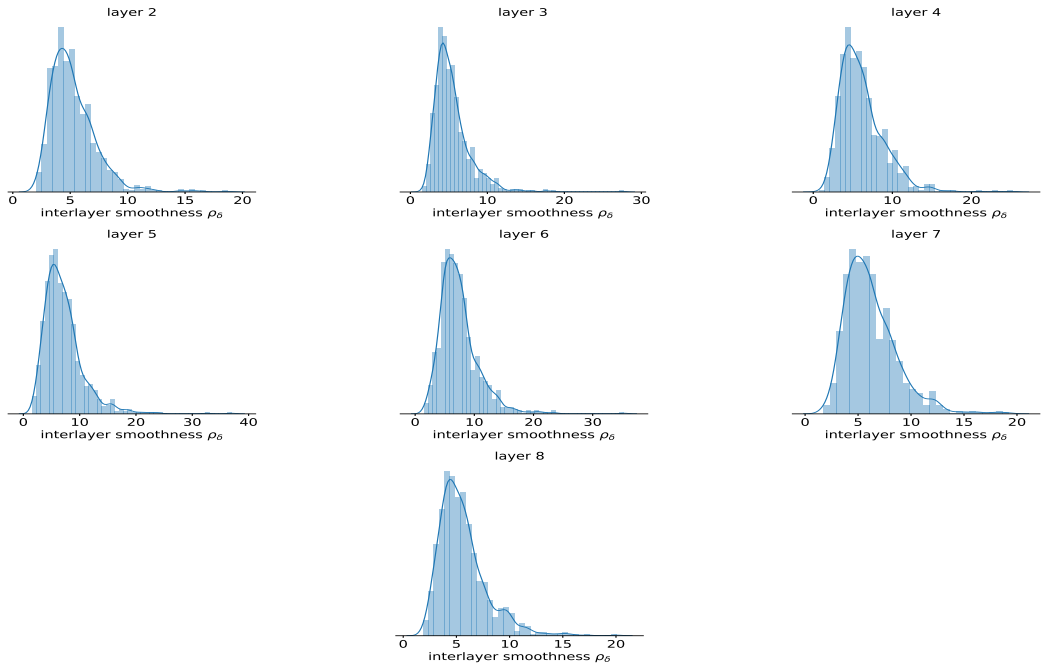
D.3.2 Interlayer cushion



D.3.3 Activation contraction



D.3.4 Interlayer smoothness



E Tools

We use matrix concentration bounds to bound the noise produced by dropping out one single layer (Lemma 5).

Lemma 7 (Matrix Bernstein; Theorem 1.6 in (Tropp, 2012))**.** *Consider a finite sequence $\{Z_k\}$ of independent, random matrices with dimension $d_1 \times d_2$. Assume that each random matrix satisfies*

$$\mathbb{E}[Z_k] = 0 \text{ and } \|Z_k\| \leq R \text{ almost surely.}$$

Define

$$\sigma^2 := \max\Big\{\|\sum_k \mathbb{E}[Z_k Z_k^*]\|, \|\sum_k \mathbb{E}[Z_k^* Z_k]\|\Big\}.$$

Then, for all $t \geq 0$,

$$\Pr\Big\{\|\sum_k Z_k\| \geq t\Big\} \leq (d_1 + d_2)\exp\Big(\frac{-t^2/2}{\sigma^2 + Rt/3}\Big).$$

As a corollary, we have:

Lemma 8 (Bernstein Inequality: Vector Case)**.** *Consider a finite sequence $\{v_k\}$ of independent, random vectors with dimension d. Assume that each random vector satisfies*

$$\|v_k - \mathbb{E}[v_k]\| \leq R \text{ almost surely.}$$

Define

$$\sigma^2 := \sum_k \mathbb{E}\big[\|v_k - \mathbb{E}[v_k]\|^2\big].$$

Then, for all $t \geq 0$,

$$\Pr\Big\{\|\sum_k (v_k - \mathbb{E}[v_k])\| \geq t\Big\} \leq (d+1)\cdot\exp\Big(\frac{-t^2/2}{\sigma^2 + Rt/3}\Big).$$